Decode X 2026 Finale

Case GRIDSHIELD

Constrained Financial Optimization Under Regulatory Shock

CodeXplorers | Case 02

Executive Summary

- Developed cost-sensitive 2-day ahead load forecasting model.
- Incorporated weather, event, lag and rolling features.
- Stage 2 shock increased peak under-forecast penalty from **₹4 → ₹6**.
- Peak asymmetry shifted from **2:1 to 3:1**.
- Stage 3 reformulated problem into constrained financial optimization.
- Delivered penalty reduction while satisfying all regulatory constraints.

Business Context & Regulatory Framework

- 15-min resolution forecast under ABT regulation.
- Under-forecast = grid drawal risk.
- Over-forecast = procurement inefficiency.
- Asymmetric penalty: **₹4 vs ₹2**.

Objective:

Minimize financial exposure, not RMSE.

Data & Feature Engineering

Data Sources

- Historical Load (2013–2021)
- Weather drivers (Temperature, Humidity, Heat Index, Rainfall, Cool Factor)
- Event indicators

Feature Engineering

- Lag features (1, 4, 96, 672 intervals)
- Rolling mean & volatility
- Cyclical encoding for seasonality

Data Integrity

- Strict chronological split
- Leakage controls

Stage 1 Backtest Results

Validation (2020)

Controlled Bias

+0.35%

Positive bias minimizes total penalty

95th Percentile Deviation

≈ 22 kW

Peak deviation controlled

Improvement vs Baseline

86% Reduction

vs naive seasonal baseline

Key Insight:

Forecasting reframed as financial risk optimization.

Regulatory Shock & Structural Drift

Regulatory Changes

Peak Under-forecast Penalty

₹4 → **₹6**

Penalty Asymmetry

2:1 → **3:1**

2021 Test Data Characteristics:

- Elevated volatility
- Higher intra-day fluctuation
- Amplified peak variability

Implication:

Peak-hour underestimation now materially more expensive.

Stage 2 Baseline Impact

(Before Recalibration)

Total Penalty	Peak Penalty	Off-peak Penalty
₹94.6L	₹18.3L	₹76.3L

Observation:

Uniform bias insufficient under new asymmetry.

High peak penalty exposure due to increased regulatory costs

Recalibration Logic

- Introduced peak-aware bias adjustment.
- Structured grid optimization.
- No model retraining.
- Shift toward higher forecast quantile during peak (~75th percentile).

Strategic Objective:

Reduce high-cost peak under-forecast risk.

Optimization approach focuses on financial outcome rather than statistical accuracy

Stage 2 Post-Recalibration Outcome

Total Penalty

₹35.7L

↓ from ₹94.6L

Peak Penalty

₹4.4L

↓ from ₹18.3L

Off-peak Penalty

₹31.3L

↓ from ₹76.3L

Total Exposure Reduction

~60%

Peak Exposure Reduction

~75%

Trade-off:

Controlled over-forecast inefficiency accepted.

Stage 3 Board Constraints

Binding Requirements:

- Total penalty ≤ Cap
- Peak underestimation >5% allowed ≤ 3 intervals
- Bias between -2% and +3%
- Average uplift ≤ 3%
- Full risk transparency

> Problem transformed into **constrained optimization.**

Reformulated Optimization Problem

Minimize:

Total deviation penalty

Subject to:

- Reliability constraint
- Bias bound
- Buffering limit
- Regulatory cap

Positioning:

Not aggressive minimization — **disciplined optimization**.

Stage 3 Strategic Adjustment

- Moderate peak-aware bias.
- Controlled off-peak uplift.
- Target bias centered near +2%.
- Average uplift maintained under 3%.
- Peak violations limited to ≤ 3.

> Balance Achieved Between:
>
> Financial prudence and reliability compliance.

Final Stage 3 Performance Summary

Reported Metrics:

Total penalty
(under cap)

Peak penalty
Minimized

Bias
[-2%, +3%] ✓

Uplift
≤ 3% ✓

Peak violations
≤ 3 ✓

95th percentile deviation
Reported

- Worst 5 deviation intervals analyzed

Risk & Sensitivity Analysis

- 95th percentile deviation reported.
- Peak volatility contribution quantified.
- Worst 5 intervals examined.
- Stress-tested under peak amplification scenarios.
- Transparent trade-off articulation.

> Complete visibility into risk exposure and mitigation strategies.

Final Board Recommendation

By transitioning from accuracy-driven forecasting to constrained financial optimization, Lumina Energy achieves:

- Regulatory compliance
- Financial prudence
- Peak-hour reliability
- Transparent governance
- Adaptive resilience under structural shifts